Knowledge Distillation Across Teacher Architectures for Edge-Efficient Plant Disease Classification

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***Abstract*—Accurate image-based plant disease classification supports precision agriculture, but high-capacity networks are inconvenient for edge or mobile deployment. This work studies knowledge distillation from large teachers into a compact MobileNetV3-Small student, asking in particular whether the *architecture of the teacher* affects the distilled student. As a preliminary demonstration on the 38-class PlantVillage dataset, a ResNet50 teacher distills into a student reaching $99.48 \pm 0.05\%$ accuracy with 1.56M parameters, about 15 times fewer than the teacher, and with a twelvefold smaller seed-to-seed standard deviation than the same architecture trained supervised — so distillation stabilizes the compact student as well as improving it. The main study uses the smaller, class-imbalanced Cassava leaf-disease dataset (five classes) under stratified five-fold cross-validation: three teachers from different architectural families — ResNet50 (residual CNN), EfficientNet-B2 (compound-scaled CNN), and ViT-B/16 (vision transformer) — are each retrained per fold and distilled into the same student. Distillation improves the student over a supervised baseline by 1.4–1.8 accuracy points, but the choice of teacher has little effect: the three students agree to within 0.5 points despite the teachers differing widely in accuracy (82.4–85.1%) and size (7.7M–85.8M parameters). The benefit of distillation thus appears to come from the procedure itself rather than from the capacity or architecture of the teacher. Compressing the student further with post-training int8 quantization, however, fails. Ten students spanning five seeds and both training objectives all collapse to 13–49%, far below the 62% majority-class rate, and the failure resists every mitigation we applied, while the same pipeline quantizes the ResNet50 teacher losslessly. Quantizing the student one block at a time shows why no mitigation helps: no single layer is at fault — the worst costs 0.2 points alone — yet quantizing all of them costs 49. The architecture that makes the student attractive at the edge is also what makes it resist quantization, so quantization-aware training is a prerequisite rather than a refinement.**

***Index Terms*—Knowledge distillation, plant disease classification, MobileNetV3, post-training quantization, cross-validation, vision transformer, edge inference**

I. Introduction

Plant disease recognition is a practical computer vision problem with direct relevance to crop monitoring and precision agriculture, a domain where deep learning has been adopted broadly for disease detection, identification, and yield-related tasks [1]. Deep neural networks can achieve strong classification accuracy on curated leaf image datasets, but models with high parameter counts may be inconvenient for mobile or edge-oriented deployment, creating a tension between predictive accuracy and computational efficiency. A range of model compression techniques — including parameter pruning, quantization, and knowledge distillation — has been developed to ease this tension [2], and efficient architecture families such as the MobileNet line [3] were designed specifically for the resulting mobile and edge constraints.

This work studies that trade-off through knowledge distillation, in which a high-capacity teacher network guides the training of a lightweight MobileNetV3-Small student. Distillation is one of several routes to a smaller deployable model alongside post-hoc weight pruning and quantization [4], but it differs from those methods in shaping the student's training directly rather than compressing an already-trained network; we revisit this distinction in Section V when we combine distillation with post-training quantization. The aim is not to claim deployment on a specific device, but to evaluate whether a compact student can retain a teacher's predictive behavior while using far fewer parameters. Beyond the standard teacher-to-student comparison, we focus on a question that a single experiment cannot answer: does the *choice of teacher architecture* change how well the student learns?

We make three contributions: (i) a preliminary single-split study on 38-class PlantVillage confirming that distillation improves a compact student; (ii) a cross-validated study on Cassava comparing three teacher architectures (ResNet50, EfficientNet-B2, ViT-B/16) under stratified five-fold cross-validation, with the teacher retrained per fold and paired significance tests against the baseline; and (iii) a post-training quantization study showing that the distilled student — unlike its teacher — cannot be quantized to int8 without collapsing, that this is a property of the architecture rather than of distillation or initialization, and that its post-quantization accuracy

Code and configurations: https://github.com/bakiceltik/knowledge-distillation-edge is so sensitive to the calibration draw that single-number int8 results for this class of architecture are unsound.

II. Related Work

PlantVillage [5] provides a widely used benchmark of labeled plant leaf images. Its clean imaging conditions are well suited to initial controlled experiments but less representative of field conditions. The Cassava leaf-disease dataset [6] covers five classes with strong class imbalance and real field imagery, providing a harder and more realistic testbed. Early work demonstrated high accuracy on PlantVillage with convolutional networks [7], but the literature has paid comparatively little attention to compressing these models for on-device inference.

Compressing a network for constrained hardware can be approached in several ways — pruning, quantization, low-rank factorization, and knowledge distillation — as surveyed by Cheng et al. [2]. This work examines the last two in sequence: we distil into a compact student and then attempt to quantize it, and we compare against pruning in Section V-F. Whether the two actually compose is not a foregone conclusion, and it turns out to be one of our findings (Section V-C).

Knowledge distillation [8] transfers information from a larger teacher to a smaller student by having the student match the teacher's softened output probabilities, which encode inter-class similarity absent from one-hot labels. Later methods go beyond output logits and transfer intermediate representations, such as hidden-layer hints [9] and spatial attention maps [10]; a broad survey is given by Gou et al. [11]. We use the original logit-based formulation since our focus is the teacher architecture rather than the distillation mechanism, and a logit-based loss is the one formulation that applies unchanged across teachers whose internal representations are not comparable — a CNN feature map and a transformer patch-token sequence have no natural correspondence.

A recurring finding is that stronger teachers do not necessarily produce better students. Cho and Hariharan [12] show that a large capacity gap can impair distillation, and Mirzadeh et al. [13] propose intermediate teacher assistants to bridge it. Beyer et al. [14] argue that training consistency matters more than teacher accuracy. These results motivate our central question: when the student is fixed, does the teacher's *architecture* — and not merely its accuracy — affect the distilled student? DeiT [15] distills *into* a transformer using a CNN teacher; our setting is complementary, distilling *from* a transformer and two CNN teachers into a convolutional student.

The three teachers span distinct architectural families: ResNet50 uses residual connections to ease optimization of deeper convolutional models [16]; EfficientNet-B2 applies compound scaling of depth, width, and resolution [17]; and ViT-B/16 processes images as patch sequences via self-attention [18]. The student in all experiments is MobileNetV3-Small [19], a compact architecture designed for mobile inference.

III. Methodology

Let z_t and z_s denote the teacher and student logits for an input image, and let y denote the ground-truth class label. Supervised teacher and supervised student baselines are trained with cross-entropy against the hard labels. For distillation, the teacher is first trained and then frozen, and the student is optimized using a weighted sum of the cross-entropy loss and a temperature-scaled Kullback–Leibler divergence term:

$$\begin{aligned}\mathcal{L} = \alpha T^2 \mathrm{KL}\Big(\mathrm{softmax}\Big(\frac{z_t}{T}\Big) \,\Big\|\, \mathrm{softmax}\Big(\frac{z_s}{T}\Big)\Big) \\ + (1-\alpha)\,\mathrm{CE}(y, z_s).\end{aligned} \quad (1)$$

The T^2 factor compensates for the gradient scaling introduced by softened probabilities. The temperature is fixed at $T = 4.0$ throughout; the distillation weight is $\alpha = 0.5$ in the PlantVillage study and $\alpha = 0.7$ in the main Cassava study. In both cases α is held constant across teachers so that any difference between teachers is attributable to the teacher rather than to the loss weighting.

IV. Experimental Setup

All experiments use pretrained ImageNet weights, images resized to 224×224 pixels and normalized with ImageNet statistics, random horizontal flipping and small rotations during training, deterministic resizing for validation and test, and the best validation checkpoint for evaluation. All training runs, on both datasets, use a single NVIDIA RTX 4070 (CUDA) with automatic mixed precision, so results are comparable across the two studies. Inference latency is measured separately, in Section V-E, on an otherwise idle machine; timings taken alongside training jobs proved worthless, and we describe the protocol there rather than reuse numbers collected during the accuracy runs.

A. Main Study: Cassava Cross-Validation

The main study uses the Cassava leaf-disease dataset (five classes) under stratified five-fold cross-validation. For each fold, the teacher is retrained from scratch on that fold's training partition and then frozen before distilling the student; this avoids the bias that would arise if a single teacher had seen images that later appear in the test fold. The fold partition is fixed by seed, so every model sees identical folds. Three teachers are evaluated — ResNet50, EfficientNet-B2, and ViT-B/16 — each distilled into MobileNetV3-Small and compared against a supervised baseline; per-fold training schedules are given in Table I. Results are reported as mean $\pm$ standard deviation across the five folds, and each distilled student is compared against the baseline with a paired two-sided t-test on per-fold accuracies. All runs use an NVIDIA RTX 4070 with automatic mixed precision.

The auxiliary analyses that require one fixed model rather than five — the post-training quantization study and the per-class error analysis — use a single 70/15/15 train/validation/test split of Cassava (14,979/3,209/3,209 images, seed 42) rather than the cross-validation folds.

TABLE I
PER-FOLD TRAINING CONFIGURATION FOR THE CASSAVA CROSS-VALIDATION STUDY. ALL TEACHERS ARE DISTILLED INTO MOBILENETV3-SMALL WITH $T = 4.0$ AND $\alpha = 0.7$.

Model	Role	Learning rate	Epochs	Batch
ResNet50	Teacher	3×10^{-4}	10	64
EfficientNet-B2	Teacher	3×10^{-4}	10	64
ViT-B/16	Teacher	1×10^{-4}	10	32
MobileNetV3-Small	Student / baseline	5×10^{-4}	15	64

On that split we retrain the supervised and distilled students under five training seeds each, varying only weight initialization and batch order. Both arms take exactly the budget used by the cross-validation student (5×10^{-4}, 15 epochs, batch 64), so the only difference between them is the loss, and the teacher — trained on the fixed split — remains valid for every replicate. *Every* single-model analysis in this paper uses one of these students (the seed-42 distilled replicate, float32 accuracy 84.51%), so the quantization study, the error analysis, and the seed-replicate comparison all refer to models trained under an identical schedule. This matters more than it may appear: an earlier version of this work quantized a student trained for only 10 epochs, whose float32 accuracy differed by 0.1 points but whose int8 accuracy differed by more than ten, and no reader could have detected the substitution from the float32 numbers alone.

B. Preliminary Study: PlantVillage Single Split

As a preliminary demonstration, a ResNet50 teacher, a supervised MobileNetV3-Small baseline, and a distilled student are trained on all 38 classes of PlantVillage. We take the training partition of the public release ($43{,}444$ images) and re-split it into our own 70/15/15 train/validation/test partition (30,412/6,516/6,516 images, seed 42), so that the test images are held out from our own training rather than inherited from the distributed split. Training uses 15 epochs, batch size 32, $T = 4.0$, and $\alpha = 0.5$. Learning rates are 3×10^{-4} (teacher), 10^{-3} (supervised student), and 5×10^{-4} (distilled student). Both students are trained under three seeds, with the split held fixed so that only initialization and batch order vary; the teacher is trained once and reused, as it is unaffected by the student's seed.

V. RESULTS

Experiments cover two datasets. On PlantVillage (38 classes, single split), we train one ResNet50 teacher and, under three training seeds each, a supervised MobileNetV3-Small baseline and a distilled student. On Cassava (five classes, five-fold CV), we train three teacher baselines, one student baseline, and three distilled students — one per teacher — plus an ablation over distillation hyperparameters and four post-training quantization variants. A separate batch-one latency benchmark (Section V-E) times every architecture on an idle machine. All metrics are macro-averaged precision, recall, and F1 alongside accuracy.

A. Cassava Cross-Validation

Table II summarizes the cross-validated results.

Distillation helps. All three teachers raise the student above the supervised baseline (81.67% accuracy) by a similar margin: $+1.82$ points for ResNet50 (83.49%), $+1.45$ for ViT-B/16 (83.12%), and $+1.37$ for EfficientNet-B2 (83.04%). In the paired t-test the ResNet50 and ViT-B/16 gains are statistically significant ($p = 0.029$ and $p = 0.036$); the EfficientNet-B2 gain does not reach significance ($p = 0.079$), most likely reflecting the limited power of five folds together with the larger fold-to-fold variance of its paired differences against the baseline (SD 1.31 points, versus 1.23 for ResNet50 and 1.05 for ViT-B/16).

The more notable result is how little the teacher matters. The three distilled students lie within 0.5 points of one another (83.04–83.49%), a spread comparable to their across-fold standard deviations (0.31–0.72), even though the teachers differ substantially in accuracy (82.40–85.13%), architectural family (two CNNs and one transformer), and size (7.7M, 23.5M, 85.8M parameters). The ViT-B/16 student (83.12%) surpasses its own teacher (82.40%), which runs counter to the expectation that a very large teacher, separated from a 1.5M-parameter student by a wide capacity gap, would distill poorly. We interpret this teacher-agnostic behavior in Section VII.

B. Sensitivity to Distillation Hyperparameters

The main study fixes $T = 4.0$ and $\alpha = 0.7$ across teachers. To verify this choice is not an artifact, we run a single-split sensitivity analysis for the ResNet50 teacher on fold 0, sweeping $T \in \{1, 2, 4, 8\}$ at $\alpha = 0.7$ and $\alpha \in \{0.3, 0.5, 0.7, 0.9\}$ at $T = 4.0$. Because T and α enter only the student loss, the teacher is trained once and reused for every setting, so the sweep varies the distillation schedule alone. Table III reports the result.

The student is stable: accuracy stays within about 1.4 points over all settings (82.78–84.14%), with a gentle maximum at the default $T = 4.0$, $\alpha = 0.7$. Setting $\alpha = 0.9$ lowers macro-F1 to 69.39%, consistent with over-weighting the teacher at the expense of minority classes. The flat response confirms that the fixed configuration is well justified. These are single-split point estimates without fold error bars, and are intended as a sensitivity check rather than as a second cross-validated comparison.

C. Edge Compression: Quantization

We apply int8 post-training quantization (PTQ) on CPU to the distilled student described in Section IV — the seed-42 replicate, trained under the same schedule as every other model in this paper. *Dynamic* PTQ is applied in eager mode; *static* PTQ (weights and activations) uses the PT2E export-based API with the XNNPACK quantizer, calibrated on 1,024 training images. Because the calibration draw is itself a nuisance factor of the size we are trying to measure, each static configuration is run over five calibration seeds and reported as mean $\pm$ SD rather than as a single number. Table IV reports the outcome.

TABLE II
CROSS-VALIDATED TEST RESULTS ON CASSAVA (5-FOLD, MEAN ± STANDARD DEVIATION ACROSS FOLDS). PRECISION, RECALL, AND F1 ARE MACRO-AVERAGED. THE TEACHER IS RETRAINED PER FOLD; THE STUDENT AND KD HYPERPARAMETERS ARE FIXED.

Teacher	Student	Accuracy (%)	Precision (%)	Recall (%)	Macro-F1 (%)	Parameters
–	resnet50	85.13 ± 0.95	75.08 ± 1.40	73.69 ± 1.28	74.00 ± 0.78	23,518,277
–	efficientnet_b2	84.59 ± 0.67	73.91 ± 1.51	71.75 ± 0.86	72.60 ± 0.61	7,708,039
resnet50	mobilenet_v3_small	83.49 ± 0.31	72.70 ± 0.66	69.74 ± 2.62	70.72 ± 1.99	1,522,981
vit_b_16	mobilenet_v3_small	83.12 ± 0.72	72.90 ± 0.99	68.25 ± 2.36	69.85 ± 1.95	1,522,981
efficientnet_b2	mobilenet_v3_small	83.04 ± 0.41	71.75 ± 1.53	69.40 ± 1.98	70.34 ± 1.26	1,522,981
–	vit_b_16	82.40 ± 0.64	71.41 ± 1.28	69.08 ± 1.58	69.38 ± 1.26	85,802,501
–	mobilenet_v3_small	81.67 ± 1.01	69.83 ± 2.64	67.57 ± 1.38	68.22 ± 0.80	1,522,981

TABLE III
SINGLE-SPLIT KD SENSITIVITY ANALYSIS FOR THE RESNET50 TEACHER (FOLD 0 OF THE CASSAVA CROSS-VALIDATION SPLIT). TEMPERATURE IS SWEPT AT $\alpha = 0.7$ AND α AT $T = 4$; † MARKS THE DEFAULT USED IN THE MAIN STUDY. PRECISION, RECALL, AND F1 ARE MACRO-AVERAGED. VALUES ARE SINGLE-SPLIT POINT ESTIMATES (NO FOLD ERROR BARS).

Sweep	T	α	Accuracy (%)	Precision (%)	Recall (%)	Macro-F1 (%)
temperature	1	0.7	83.50	73.04	70.05	71.35
temperature	2	0.7	83.79	74.38	69.01	71.29
temperature	4	$0.7^{\dagger}$	84.14	73.99	70.29	71.52
temperature	8	0.7	83.67	73.26	70.03	71.37
alpha	4	0.3	82.78	70.59	70.76	70.61
alpha	4	0.5	83.60	72.91	69.64	70.89
alpha	4	$0.7^{\dagger}$	84.14	73.99	70.29	71.52
alpha	4	0.9	83.11	73.01	67.04	69.39

Two properties of the tooling must be stated, because both are easy to misreport. First, PyTorch's eager-mode dynamic quantization supports Linear and recurrent layers only: on a convolutional network it converts the classifier head and silently leaves every convolution in float32. Our "dynamic int8" row is therefore an int8 classifier head atop a float32 backbone — 2 Linear layers quantized, all 52 convolutions untouched — and we label it as such rather than as an int8 model. Second, the PT2E static path emits a reference-quantized graph that simulates int8 arithmetic with explicit quantize/dequantize operators in float. Its accuracy faithfully reflects int8 inference, but its serialized size and runtime do not, so we report no size or latency for the static rows.[1]

Dynamic quantization is essentially lossless ($84.51 \rightarrow 84.48\%$) and shrinks the model $1.4\times$, but the caveat above explains why: it never touches the convolutional backbone, so the reduction is confined to the classifier head and the invariance of accuracy is unsurprising rather than a result. It is not a route to an int8 edge model. It also yields no batch-one CPU speed-up (Section V-E: 16.3 ms either way), because the classifier head was never where the time was going. Size and latency are distinct axes, and quantizing the head moves only the first. Our measurements use the `fbgemm` x86 backend; an ARM backend (`qnnpack`) would be the target for actual mobile deployment.

Static quantization, which does quantize the convolutions, collapses the student from 84.5% to $35.5 \pm 7.6\%$ — far below the 62% obtained by always predicting the majority class, and therefore unusable. Four controls establish that this is a genuine property of the model rather than an artifact of our configuration, and each rules out a candidate explanation.

It is not calibration-limited. Enlarging the calibration set eightfold leaves the student in the same collapsed regime: $28.6 \pm 6.0\%$ at 256 images, $35.5 \pm 7.6\%$ at $1{,}024$, and $33.0 \pm 5.8\%$ at $2{,}048$. The spread between those means (6.9 points) is no larger than the draw-to-draw noise within a single size (≈ 6.5 points), so we cannot even resolve a size effect — and more to the point, no amount of calibration data rescues the model. We stress that this conclusion requires several draws per size: a single draw per size would have produced a 17-point spread that is pure noise, and would have invited exactly the wrong inference.

It is not the squeeze-and-excitation blocks. Holding every SE block in float32 while still quantizing the convolutions changes nothing: $35.97 \pm 7.60\%$ against $35.48 \pm 7.63\%$, a difference an order of magnitude smaller than the noise on either. The SE blocks are the usual suspect for MobileNetV3 quantization failures, and here they are exonerated.

It is not activation dynamic range. The peak absolute activation entering a convolution is 535 in ResNet50 and 105 in MobileNetV3-Small — yet ResNet50 quantizes losslessly (below) and the student does not. The model with the *wider*

[1] We initially obtained the static results through the legacy FX graph-mode API. That path is unsound for this model on torch 2.6: converting with the quantization configuration disabled everywhere — that is, quantizing nothing — still collapses test accuracy from 84.2% to roughly 15%, so it measures a defect in the conversion pass rather than the effect of quantization. All static numbers reported here use the PT2E API.

TABLE IV
POST-TRAINING QUANTIZATION OF THE DISTILLED MOBILENETV3-SMALL ON THE CASSAVA SINGLE-SPLIT TEST SET. PRECISION, RECALL, AND F1 ARE MACRO-AVERAGED. STATIC ROWS USE PT2E REFERENCE QUANTIZATION, WHOSE SIMULATED GRAPH MAKES SERIALIZED SIZE UNREPRESENTATIVE OF A DEPLOYED INT8 MODEL (–). CHANCE IS 20% AND THE MAJORITY-CLASS RATE IS 62%. STATIC INT8 ACCURACIES ARE MEAN $\pm$ SD OVER *five calibration draws*: A SINGLE DRAW IS NOT A MEASUREMENT ON THIS ARCHITECTURE, SINCE THE DRAW ALONE MOVES ACCURACY BY SEVERAL POINTS.

Variant	Accuracy (%)	Precision (%)	Recall (%)	Macro-F1 (%)	Size (MB)
Float32 (baseline)	84.51	74.77	70.97	72.57	5.93
Dynamic int8 (classifier head only; 0/52 convs)	84.48	74.62	70.88	72.45	4.23
Static int8, per-channel weights	35.48 ± 7.63	47.21	29.14	20.25	–
Static int8, per-channel, SE blocks in float32	35.97 ± 7.60	46.57	29.44	20.79	–
Static int8, per-tensor weights	10.10 ± 0.24	12.72	19.66	7.26	–

TABLE V
PER-LAYER SENSITIVITY: EACH ROW QUANTIZES *only* THAT BLOCK-GROUP AND LEAVES THE REST OF THE NETWORK IN FLOAT32. NO SINGLE GROUP REPRODUCES THE COLLAPSE; QUANTIZING ALL OF THEM TOGETHER COSTS 49 POINTS. THE FIVE MOST DAMAGING GROUPS ARE SHOWN; THE REMAINING NINE COST < 0.07 POINTS EACH.

Quantized group	Accuracy (%)	Drop (pts)
None (float32)	84.51	—
`features.5` only	84.29	0.22
`features.6` only	84.33	0.19
`features.8` only	84.39	0.12
`features.3` only	84.42	0.09
`classifier` only	84.45	0.06
All groups	$\mathbf{35.48 \pm 7.63}$	$\mathbf{49.0}$

range is the one that survives, which inverts the obvious hypothesis.

What does matter is weight granularity. Per-tensor quantization drops the student to $10.1 \pm 0.2\%$ — *below* the 20% chance rate, because the model degenerates into predicting essentially one class for every image. So per-channel weights are necessary, but nowhere near sufficient.

The failure is distributed, not localized. We quantize the student one block-group at a time, leaving everything else in float32 (Table V). No single group matters: the worst of the fourteen costs 0.22 points, and most cost nothing measurable. Yet quantizing all of them together costs 49 points. The collapse is therefore not the fault of any one layer that could be exempted — it is an accumulation of individually harmless errors across the depthwise-separable stack. This explains why exempting a module type (the SE control above) buys nothing, and why quantization-aware training, which lets the network compensate for the accumulated error during training, is the appropriate remedy rather than a better post-hoc configuration.

The decisive question is whether the collapse is a property of *our* student — its distillation schedule, or the particular optimum it converged to — or of the MobileNetV3-Small architecture itself. To settle it we replicate both student arms over five training seeds. The data split is held fixed, so the teacher never sees a test image and only the student's initialization and batch order change, and both arms use identical budgets (5×10^{-4}, 15 epochs, batch 64): the sole difference between them is the loss. Each of the ten resulting students is quantized through the same PT2E path. Table VI reports the outcome.

The cause is the architecture. Every one of the ten students collapses: int8 accuracy spans 12.8–49.4% against a float32 range of 81.5–84.8%, and not one comes close to the 62% majority-class rate. The supervised students — never distilled, and therefore different initializations converging to different optima — collapse just as the distilled ones do. Neither distillation nor the student's initialization can account for the failure. Meanwhile the ResNet50 teacher passes through the identical pipeline losslessly: over five calibration draws it scores $86.65 \pm 0.17\%$ against a float32 86.51%, a difference indistinguishable from zero. This confirms both that the quantization path is sound and that the failure is specific to the compact architecture.

The replication further shows that post-quantization accuracy is too unstable to support any finer claim, and we report this explicitly because it governs how the int8 numbers in this paper should be read. Two independent sources of noise act on it. Across training seeds, int8 accuracy varies with a standard deviation of 9.0 points (supervised) and 13.2 points (distilled), even though the corresponding float32 accuracies vary by less than a point. On top of that, the calibration draw alone moves the result: holding a single checkpoint completely fixed and varying only which 1,024 augmented training images are used for calibration, the distilled student's int8 accuracy ranges over 27.6–44.2% (35.5 ± 7.6). A single int8 figure is therefore one sample from a wide distribution over two nuisance factors, and we fix both the training and calibration seeds so that the values we report reproduce.

This instability is itself architecture-specific, and the contrast is stark. Subjected to the same five calibration draws, ResNet50's int8 accuracy has a standard deviation of 0.17 points against the student's 7.63 — a factor of forty-five. MobileNetV3-Small does not merely quantize badly; the very *measurement* of how badly it quantizes is unstable, whereas for ResNet50 post-training quantization is both lossless and repeatable. Sensitivity to the calibration set is thus a second symptom of the same underlying cause, and it suggests that reporting int8 accuracy for depthwise-separable architectures without quantifying calibration variance is unsound practice.

TABLE VI

IS THE INT8 COLLAPSE CAUSED BY OUR STUDENT, OR BY ITS ARCHITECTURE? THE TWO MOBILENETV3-SMALL ARMS ARE REPLICATED OVER FIVE TRAINING SEEDS ON A FIXED SPLIT WITH MATCHED TRAINING BUDGETS, SO ONLY THE LOSS DIFFERS. EVERY MODEL HERE — STUDENTS AND TEACHER ALIKE — IS QUANTIZED OVER THE *same* FIVE CALIBRATION DRAWS AND AVERAGED, SO THE STUDENT SDS ARE VARIATION ACROSS TRAINING SEEDS AND NOT CALIBRATION NOISE; THE TEACHER IS A SINGLE CHECKPOINT, SO ITS SD IS CALIBRATION NOISE ALONE. ALL MODELS ARE QUANTIZED THROUGH THE IDENTICAL PT2E PATH (PER-CHANNEL WEIGHTS, 1,024 CALIBRATION IMAGES) AND EVALUATED ON THE CASSAVA SINGLE-SPLIT TEST SET. THE INT8 STANDARD DEVIATIONS ARE LARGE BY NATURE, NOT BY ACCIDENT: SEE THE TEXT ON CALIBRATION NOISE.

Model	Float32 (%)	Int8 (%)	Paired p (int8)
MobileNetV3-Small, supervised (5 seeds)	82.66 ± 0.66	24.32 ± 8.99	
MobileNetV3-Small, distilled (5 seeds)	84.32 ± 0.36	29.86 ± 13.21	0.52 (n.s.)
ResNet50, teacher (5 calib. draws)	86.51	86.65 ± 0.17	–

This has a direct consequence. On one seed the distilled student retained far more accuracy under quantization than the supervised one, which invites the conclusion that distillation confers quantization robustness. Over five paired seeds that gap is $+5.5 \pm 17.5$ points and not significant (paired t-test, $p = 0.52$); across repeated measurements its point estimate wandered between $+1.6$ and $+10.0$ points without ever approaching significance. **We therefore make no claim that distillation aids quantization.** By contrast, the same replication reproduces the *float32* benefit of distillation exactly and stably ($+1.66 \pm 0.95$ points, $p = 0.017$) under a fixed split and a matched budget, corroborating the cross-validated result of Section V-A by an independent route. The contrast is instructive: distillation's effect on float accuracy is small but highly reproducible, whereas its apparent effect on quantized accuracy is large but pure noise.

All of this is consistent with MobileNetV3-Small being the one architecture in its family for which `torchvision` ships no post-training quantized variant, while shipping one for ResNet50 and for MobileNetV3-Large. Recovering int8 accuracy on this student requires quantization-aware training, which we leave to future work.

D. PlantVillage Single Split

Table VII reports the results on the 38-class PlantVillage task. Both students are replicated over three training seeds on the fixed split, which matters here: a single run of this experiment is misleading, and in a way that happens to flatter neither conclusion.

Distillation raises the student from $97.84 \pm 0.59\%$ to $99.48 \pm 0.05\%$, a paired gain of $+1.64 \pm 0.64$ points ($p = 0.047$). The distilled student not only overtakes its supervised twin — same architecture, same parameter count, same budget, differing only in the loss — but matches the ResNet50 teacher itself (99.48%) at a fifteenth of its size.

The more interesting number is the standard deviation. Across seeds the supervised baseline has an SD of 0.59 points while the distilled student has 0.05 — a twelvefold smaller standard deviation. Distillation does not merely lift the student's accuracy on this dataset, it very nearly removes its run-to-run variability, which is consistent with reading the soft-label term as a regularizer.

This asymmetry is also what makes a single run of the experiment treacherous, and the treachery is one-sided. All of the variance sits in the baseline: the three supervised seeds span 97.38–98.50%, while the three distilled seeds span 99.42–99.51%. Because the distillation gain is measured *against* that baseline, a single split estimates the gain only as well as it happens to draw the noisy cell — a lucky baseline halves the apparent gain, an unlucky one inflates it. On a near-saturated dataset the honest reading is that the direction is certain and the magnitude is not, which is why the quantitative claims of this paper rest on the cross-validated Cassava study.

E. Inference Latency

Parameter count is the compression metric used above, but it is a proxy, and the quantity that actually matters at the edge is time per image. We therefore benchmark batch-one inference on an idle machine: 50 warm-up passes (which also let cuDNN autotuning settle), then 300 timed passes, with the entire measurement repeated five times from scratch. We report the median and inter-quartile range rather than the mean, because a single scheduler outlier is enough to corrupt a mean, and we additionally report the spread between the five independent repeats. That last column is a self-check: if independent repeats of the same measurement disagree, the median is not a real number and the reader should be told so.[2] Table VIII reports the result. CPU timings use eight pinned threads; the GPU is the RTX 4070 used throughout.

Three things follow, and none of them is the one the parameter counts predict.

First, **parameter count is a poor proxy for latency**. EfficientNet-B2 has a third of ResNet50's parameters yet is nearly twice as slow on the GPU (8.25 against 4.68 ms), and ViT-B/16 has 3.6× *more* parameters than ResNet50 while running faster than EfficientNet-B2. Depth, memory traffic, and kernel-launch overhead dominate at batch one; the 15× compression we report elsewhere should not be read as a 15× speed-up.

Second, **the student's advantage is a CPU advantage**. On the GPU the distilled student is only 1.18× faster than its ResNet50 teacher (3.96 against 4.68 ms) despite being fifteen times smaller — at batch one a GPU is latency-bound by kernel launches, not by arithmetic, so a small model cannot exploit its size. On the CPU, which is what an edge device actually resembles, the same student is 6.5× faster (15.5 against 100.3 ms). The case for distilling into a compact architecture is therefore a case about CPU-class hardware, and evaluating it on a desktop GPU alone would have understated it more than fivefold.

Third, **dynamic int8 buys no speed**, exactly as its accuracy result predicted: 17.2 ms quantized against 15.5 ms in float

[2]This protocol is a response to a failure. Our first latency measurements were taken while training jobs were resident on the GPU, and two checkpoints of the *same* architecture — differing only in training seed — timed a factor of two apart. Those numbers were discarded.

TABLE VII

TEST RESULTS ON THE 38-CLASS PLANTVILLAGE TASK (SINGLE 70/15/15 SPLIT, RTX 4070). THE TWO MOBILENETV3-SMALL ROWS ARE MEAN $\pm$ SD OVER THREE TRAINING SEEDS; THE TEACHER IS A SINGLE RUN. PRECISION, RECALL, AND F1 ARE MACRO-AVERAGED. NOTE THE ORDER-OF-MAGNITUDE DIFFERENCE IN THE STUDENTS' STANDARD DEVIATIONS.

Model	Training strategy	Accuracy (%)	Precision (%)	Recall (%)	Macro-F1 (%)	Parameters
ResNet50	Teacher supervised	99.48	99.29	99.35	99.31	23,585,894
MobileNetV3-Small	Supervised baseline	97.84 ± 0.59	97.11 ± 0.53	96.78 ± 0.92	96.75 ± 0.73	1,556,806
MobileNetV3-Small	Distilled from ResNet50	99.48 ± 0.05	99.42 ± 0.08	99.19 ± 0.14	99.29 ± 0.10	1,556,806

TABLE VIII

BATCH-ONE INFERENCE LATENCY (MEDIAN, WITH INTER-QUARTILE RANGE) ON AN IDLE MACHINE. "SPREAD" IS THE DIFFERENCE BETWEEN THE MEDIANS OF FIVE INDEPENDENT REPEATS OF THE WHOLE MEASUREMENT — A RELIABILITY CHECK, NOT A RESULT: EVERY SPREAD HERE IS SMALL RELATIVE TO THE DIFFERENCES BETWEEN MODELS, SO THE MEDIANS ARE TRUSTWORTHY.

Model	Params	CPU (ms)	GPU (ms)	Spread (CPU/GPU)
ResNet50 (teacher)	23.5M	100.3	4.68	5.4 / 0.36
EfficientNet-B2 (teacher)	7.7M	46.8	8.25	2.5 / 0.14
ViT-B/16 (teacher)	85.8M	129.9	7.34	6.6 / 0.38
MobileNetV3-Small (student)	1.5M	15.5	3.96	2.4 / 0.09
+ dynamic int8 (head only)	1.5M	17.2	–	4.1 / –

— if anything slightly slower, and well within the run-to-run spread. This is the same finding from the other side: the quantization touched only the classifier head, and the classifier head was never where the time was going.

F. Comparison with the Literature

To situate our absolute numbers, Table IX places them beside published results on the same two datasets. Evaluation protocols differ — Mohanty et al. use an 80/20 split, Liu et al. a 90/10 split, and we use stratified five-fold cross-validation — so the comparison controls for the dataset, and approximately for the architecture, but not for the evaluation protocol.

On PlantVillage the distilled student reaches $99.48 \pm 0.05\%$ accuracy, above the 99.35% of Mohanty et al.'s GoogLeNet [7] and approaching the 99.75% of Too et al.'s DenseNet-121 [20]. The dataset is close to saturated at this accuracy, so the meaningful axis is model size. Pruning is the natural point of comparison: in a separate study, Too et al. [21] prune DenseNet for exactly this purpose, and their best pruned model retains 99.24% on PlantVillage while reducing parameters by about 14% and FLOPs by about 25%. Our distilled student attains a comparable 99.48% with 1.56M parameters. Relative to the unpruned DenseNet-121 ($\approx$7.0M), that is a $4.5\times$ reduction, against the $1.16\times$ that filter pruning achieves — so distilling into an architecture designed for mobile inference compresses roughly four times more aggressively than pruning a large backbone, at no cost in accuracy. The two are complementary rather than competing, and the protocols are not identical, so this is an indication rather than a controlled comparison.

The Cassava comparison is the more informative one, because Liu et al. [22] evaluate single models on the *same* Kaggle Cassava Leaf Disease Classification release [6] that we use — the same five classes, collected in Uganda by the National Crop Resources Institute and the Makerere University AI Laboratory — so their per-architecture numbers are comparable to ours. They report 85.2% for ResNet50 and 84.1% for "MobileNet v3", against our $85.13 \pm 0.95\%$ ResNet50 teacher and $83.49 \pm 0.31\%$ distilled MobileNetV3-Small student. The ResNet50 comparison is exact. The MobileNetV3 one is not: Liu et al. do not state which variant they use and report no parameter counts, and since MobileNetV3-Large is the more common default their figure may well correspond to the larger model, in which case our Small student is achieving a comparable score with roughly a third of the parameters. We therefore read this row as indicative rather than as a like-for-like match. Either way our models land within about a point of independently published results on the same data, which supports the correctness of our training pipeline and indicates that the $\sim$84% regime is simply what single models attain on this dataset rather than a shortfall of our setup. Stronger results exist: the multi-scale fusion model of Liu et al. attains 88.1%, the winning entry of the associated Kaggle competition reached 91.32% on the private leaderboard, and the best reported result we are aware of is the 91.43% of Che et al. [23]. These, however, rely on ensembles, test-time augmentation, object segmentation, and considerably longer training schedules, all of which would confound the controlled teacher-architecture comparison that is the objective of this work.

VI. ERROR ANALYSIS

The gap between accuracy and macro-F1 throughout the Cassava results is a direct consequence of class imbalance. On the single-split test set, *mosaic disease* accounts for $1{,}984$ of $3{,}209$ images (62%), whereas *bacterial blight* has only 155 (5%). Table X and Figure 1 give the per-class performance and confusion matrix of the distilled student.

The student performs well on the dominant class (mosaic disease, F1 0.94) but degrades on minority classes, reaching its worst on bacterial blight (F1 0.56, recall 0.50). Because accuracy is dominated by the majority class while macro-F1 weights all classes equally, accuracy (84.5%) substantially overstates per-class competence (macro-F1 72.6%), which is why we report both throughout. The confusion matrix reveals two systematic error patterns. First, minority diseases are frequently predicted as *healthy*: 51 of 155 bacterial-blight leaves (33%) and 38 brown-streak leaves are missed this way. These are false negatives — diseased plants reported as healthy — which is the costliest error type in crop monitoring,

TABLE IX

COMPARISON WITH PUBLISHED RESULTS ON THE SAME DATASETS. THE CASSAVA ROWS OF LIU ET AL. USE THE SAME KAGGLE RELEASE AND THE SAME FIVE CLASSES AS OURS, SO THE ARCHITECTURES ARE DIRECTLY COMPARABLE; THE EVALUATION PROTOCOL STILL DIFFERS (THEIR 90/10 SPLIT VS. OUR STRATIFIED FIVE-FOLD CROSS-VALIDATION), AS DOES MOHANTY ET AL.'S 80/20 PLANTVILLAGE SPLIT. LITERATURE VALUES ARE AS REPORTED BY THE CITED AUTHORS. OUR CASSAVA ROWS ARE MEAN $\pm$ SD OVER FIVE FOLDS; OUR PLANTVILLAGE STUDENT ROW IS MEAN $\pm$ SD OVER THREE TRAINING SEEDS ON A SINGLE SPLIT. PARAMETER COUNTS MARKED $\approx$ ARE THE STANDARD IMAGENET VARIANTS OF THE CITED ARCHITECTURES, ADJUSTED FOR THE 38-CLASS HEAD; TOO ET AL. [21] REPORT THEIR PRUNING AS A RELATIVE REDUCTION RATHER THAN AN ABSOLUTE COUNT. OUR PARAMETER COUNTS DIFFER SLIGHTLY BETWEEN THE TWO DATASETS BECAUSE THE CLASSIFIER HEAD IS SIZED TO THE NUMBER OF CLASSES (38 FOR PLANTVILLAGE, 5 FOR CASSAVA); THE BACKBONES ARE IDENTICAL.

Work	Model	Accuracy (%)	Parameters
PlantVillage (38 classes)			
Mohanty et al. [7]	GoogLeNet	99.35	$\approx$6.8M
Too et al. [20]	DenseNet-121 (unpruned)	99.75	$\approx$7.0M
Too et al. [21]	DenseNet (pruned)	99.24	$\approx$14% fewer
Ours	ResNet50 (teacher)	99.48	23.6M
Ours	MobileNetV3-Small (distilled)	99.48 ± 0.05	1.56M
Cassava (5 classes)			
Liu et al. [22]	MobileNetV3	84.1	–
Liu et al. [22]	ResNet50	85.2	–
Liu et al. [22]	EfficientNet-B6	86.5	–
Liu et al. [22]	MSFM (proposed, multi-scale)	88.1	–
Ours	ResNet50 (teacher)	85.13 ± 0.95	23.5M
Ours	MobileNetV3-Small (baseline)	81.67 ± 1.01	1.5M
Ours	MobileNetV3-Small (distilled)	83.49 ± 0.31	1.5M

TABLE X

PER-CLASS PERFORMANCE OF THE DISTILLED MOBILENETV3-SMALL ON THE CASSAVA SINGLE-SPLIT TEST SET.

Class	Precision	Recall	F1	Support
Bacterial blight	0.642	0.497	0.560	155
Brown streak disease	0.772	0.707	0.738	307
Green mottle	0.773	0.693	0.731	378
Healthy	0.626	0.701	0.662	385
Mosaic disease	0.925	0.951	0.938	1,984

and the fact that they concentrate in the rare classes means the headline accuracy conceals them almost entirely. Second, the two visually similar mottling classes, *green mottle* and *mosaic disease*, are mutually confused (67 and 29 cases), reflecting genuine visual similarity rather than imbalance alone. Distillation improves the student overall but inherits this confusion structure rather than resolving the hardest minority cases; addressing them would likely require class re-balancing, targeted augmentation, or cost-sensitive training.

A t-SNE projection of the student's penultimate embeddings (Figure 2) tells the same story geometrically: *mosaic disease* forms a large, well-separated cluster, consistent with its 0.94 F1, whereas *bacterial blight*, *green mottle*, and *healthy* overlap in a shared region — precisely the minority classes the confusion matrix shows being conflated. The representation is therefore well organized for the majority class but only weakly discriminative for the rare ones.

VII. DISCUSSION

The central observation is that the distilled student is largely insensitive to its teacher: across a residual CNN, a compound-scaled CNN, and a vision transformer — differing widely in accuracy, architecture, and size — the three students fall within

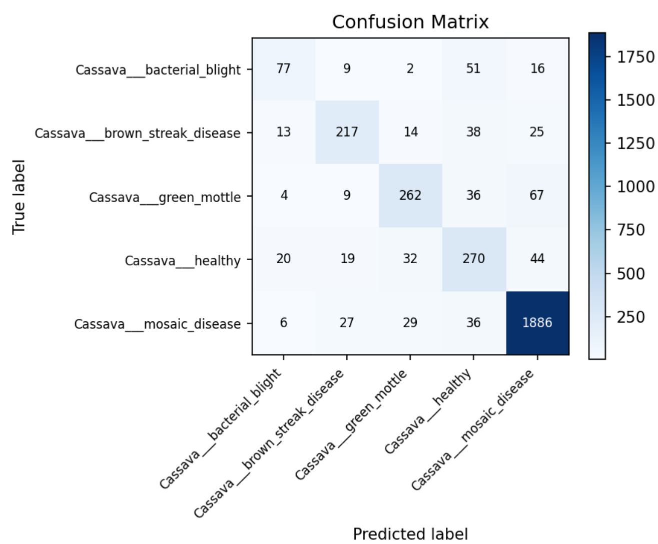

Fig. 1. Confusion matrix of the distilled MobileNetV3-Small on the Cassava single-split test set. Errors concentrate in the minority classes.

0.5 points of one another, and the weakest and largest teacher (ViT-B/16) distills as effectively as the strongest. This points to the softened-label regularization of distillation, rather than the teacher's capacity or architecture, as the source of the 1.4–1.8-point gain. The implication for edge settings is convenient: a small, inexpensive teacher suffices. The latency benchmark adds a caveat to the compression story, however. The student's $15\times$ parameter reduction converts to only a $1.17\times$ batch-one speed-up on the GPU but roughly $6.5\times$ on the CPU, and among the teachers themselves parameter count and latency are not even monotonically related. Compression ratios and speed-ups are different currencies, and papers in this area —

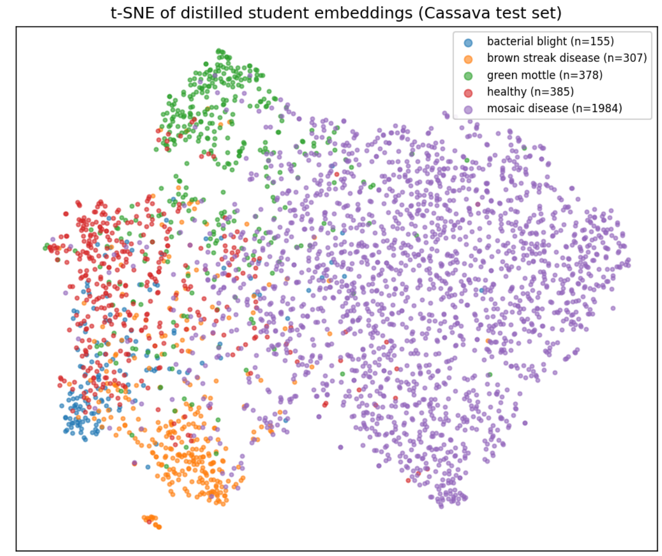

Fig. 2. t-SNE projection of the distilled student's 1024-dimensional penultimate embeddings on the Cassava single-split test set. The majority *mosaic disease* class separates cleanly, while the minority classes overlap, mirroring the per-class scores (Table X) and the confusion matrix (Figure 1).

ours included, before we measured — routinely quote the first while implying the second.

Compression, by contrast, is where the compact student proves brittle. Quantization that leaves the convolutional backbone in float is harmless but buys little, while genuine int8 quantization of the convolutions destroys the model, and no configuration we tried — calibration size, weight granularity, or exempting the squeeze-and-excitation blocks — recovers it. The cause is the architecture rather than our student: the same pipeline quantizes the ResNet50 teacher losslessly and repeatably ($86.65 \pm 0.17\%$), while ten MobileNetV3-Small students spanning five seeds and both training objectives collapse without exception. Distillation and post-training quantization are therefore not the freely composable pair that the framing of *distil, then quantize* suggests — the very design that makes MobileNetV3-Small attractive at the edge is what makes it resist PTQ. The practical reading is that the student architecture, not the teacher, is the consequential choice for a quantized deployment, which is an inversion of where this work began. We explicitly do *not* claim that distillation improves quantizability: a single pair of checkpoints suggested it did, but replication over five seeds showed the apparent advantage to be within the noise of an extremely unstable quantity ($p = 0.52$). That instability is itself worth reporting. Post-training-quantization accuracy on this architecture is not a property of the trained model alone — it also depends on which images happen to be drawn for calibration, to the tune of ± 8 points on fixed weights. Studies that report a single int8 number for a MobileNetV3-class model, ourselves included until we checked, are reporting one sample from a wide distribution.

Several limitations qualify these conclusions. Distillation hyperparameters are held fixed across teachers, so a teacher-specific T or α might widen the currently negligible gaps. With only five folds, statistical power is limited, as the borderline EfficientNet-B2 result illustrates; our claim is therefore that any teacher effect is *small*, not that it is exactly zero, and more folds or repeated runs would tighten the bound. Retraining every teacher on every fold is what licenses that claim — it is the only way to keep a teacher from having seen images that later appear in its own test fold — but it multiplies training cost by the number of folds, which is why the comparison is limited to three teachers. Finally, both datasets are curated, so accuracy on field images with cluttered backgrounds and varying lighting may be lower. Latency is measured on desktop x86 and an RTX 4070 rather than on an ARM edge device, so while the CPU-versus-GPU contrast of Section V-E is informative about *where* a compact student pays off, the absolute milliseconds are not a deployment benchmark. The PlantVillage study uses a single split rather than cross-validation; we mitigate this by replicating both students over three seeds, which is what revealed that its supervised baseline — not its distilled student — carries essentially all of the run-to-run variance on that near-saturated dataset.

VIII. Conclusion

We studied knowledge distillation from large teachers into a compact MobileNetV3-Small student for edge-efficient plant disease classification. On PlantVillage, a ResNet50 teacher distills into a student with $99.48 \pm 0.05\%$ accuracy at about $15\times$ fewer parameters, and with a twelvefold smaller seed-to-seed standard deviation than the same student trained supervised — distillation stabilizes the compact model as well as improving it. The main cross-validated study on Cassava showed that distillation raises the student by 1.4–1.8 accuracy points over a supervised baseline, yet the choice of teacher architecture — ResNet50, EfficientNet-B2, or ViT-B/16 — had little effect, indicating the benefit stems from the distillation procedure itself rather than from the teacher. This is encouraging for deployment, since a small, cheap teacher suffices. A quiesced latency benchmark tempers the compression claim: the student's $15\times$ parameter reduction yields only a $1.17\times$ batch-one speed-up on a GPU but about $6.5\times$ on a CPU, so the benefit of distilling into a compact architecture is realized on CPU-class hardware and not on the accelerator that trained it. Stacking post-training quantization on top of the distilled student, however, does not work: quantizing only the classifier head is lossless but leaves the convolutional backbone in float32, whereas int8 quantization of the convolutions collapses the student and survives every mitigation we applied. Control experiments locate the cause in the MobileNetV3-Small architecture rather than in distillation — the ResNet50 teacher quantizes losslessly through the same pipeline, while ten students spanning five seeds and both training objectives collapse without exception — so quantization-aware training is a prerequisite, not a refinement, for deploying this student in int8.

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